CURRICULUM VITAE

ER.PRATEEK AGNIHOTRI

Designation: Assistant Professor,Lecturer, Associate Faculty position

Department:Electronics& Communication Eng. ,Electrical & Electronics Eng.MASTER OF BUISNESS ADMINISTRATION[HR,MARKETING,Admission Cell, Finance , CRM,INFORMATION TECHNOLOGY]BBA Dept.Assistant Prof. Faculty , Associate Faculty staff

Qualification :B.TECH[Electronics & Telecommunication Eng.],M.TECH[VLSI & EMBEDDED SYSTEM DESIGN],M.B.A [INFORMATION TECHNOLOGY],DIPLOMA IN ELEMENTRY EDUCATION (D.EL.ED)

Phone: 7991613936,9451218659 Email: agnihotriprateek.88@gmail.com,

Skill Set

OBJECTIVE: To associate with multi-dynamic institute, which may provide a platform to update my knowledge & skills in accordance with latest trends in teaching along with research follows a tradition of anticipating & leading changes

Proficiency: (Change or Add more as necessary)

HDL	Hardware	Programming languages	Tools
Verilog	ZedBoard	C (Basic),VHDL,PYTHON,VERILOG,SYSTEM VERILOG	QuestaSim, Vivado,MATLAB,,SPICE,MySQL,Xilinx ISE9.2

Experience

- Worked with **Bharat Electronics Limited** Ghaziabad from June,2012till Dec, 2012 as Trainee Engineer [contract position]
- **Worked with Bhabha Institute of Science and Technology[BIST] LECTURER,MANAGEMENT ,ECE DEPARTMENT TECHNICALFACULTY Kanpur Dehat from July 2012to june 2017.**

Major Projects

- **Project 1**: **Project** – Functional Specification for SPI IP PROTOCOL Design
  March 2022to Aug 2022

Master mode and slave mode

• Bi-directional mode • Slave select output • Serial clock with programmable polarity and phase. Clock Polarity and Clock Phase:Data Transmission .

- **MOSI:** MOSI stands for Master Output Slave Input. It is used to send data from the master to the slave.
- **MISO:** MISO stands for Master Input Slave Output. It is used to send data from the slave to the master.
- **SCK or SCLK (Serial Clock):** It is used to the clock signal.
- **SS/CS (Slave Select / Chip Select):** It is used by the master to send data by selecting a slave.

Note: If the single slave is present in the communication, that required only three wires. The SS (slave select) is not required in in it.

a. **Project 2 – Designing of To Generate a True Rndom Number Generator using vedic multiplier, ring oscillator circuit XOR Gate , Tetra Hedral Ring Oscillation LFSR IDMA AND COMPARE CALIBRATION GENERATION METHOED USING VHDL SIMULATION**
SEP,2021 to SEP, 2022

we propose a new method for generation of TRNG using tree based interleaving technique. The output of one oscillator is send to SIPO register using a XOR gate. The other input of XOR gate is kept at '1'. The property of XOR gate is that the output bit will be independent if its input bits are independent. After the SIPO register the output is connected to the XOR based postprocessor. Among Von Neuman, BCH code and XOR based preprocessor we have used XOR based preprocessor due to ease of implementation. Finally, for implementing tree based interleaving multiplexer is used with its select lines connected to binary up down counter.

Minor Projects

- **Mini project 1: Desighning a 10 gb Ethernet Card june 2011to aug 2011**
  - Client: Ducat Noida
  - Location: Noida
  - Project: Design a 10 gb Ethernet card
  - Tools & Technology: VHDL, VERILOG SYSTEM
  -
  - **Project Description:**
  - The Ethernet specification served as the basis for the IEEE 802.3 standard, which specifies the physical and lower software layerr

Mini project 2: Project – Designing of Microcontroller 8085,8086
Feb,2010 to Mar, 2011

- Client: DUCAT Noida

- Location: Noida
- Project: Designing of Microcontroller 8085,8086
- Tools & Technology: Embedded System, VHDL, VERILOG
- **Project Description:**
- The 8085 incorporates the functions of the 8224 (clock generator) and the 8228 (system controller), increasing the level of integration. A downside compared to similar contemporary designs (such as the Z80) was the fact that the buses required de-multiplexing

Educational Qualifications

Course	Institution	University / Board	Aggregate Percentage	Batch
M.Tech. (VLSI Design & Embedded System)	HARCOURT BUTLER TECHNICAL UNIVERSITY KANPUR	H.B.T.U KANPUR UTTAR PRADESH	62.5	2022
M.BA (IINFORMATION TECHNOLOGY)	AMITY UNIVERSITY SECTOR-125 Noida UTTAR PRADESH-201303	AMITY UNIVERSITY ODL ,Noida Uttar Pradesh	65.06	2021-23
B.Tech (Electronics & Communication)	VISHVESHWARYA INSTITUTEOF ENG. & Tech. G. B NAGAR GR. NOIDAU.P	Dr.APJ ABDUL KALAM TECHNICAL UNIVERSITY LUCKNOW U.P	65.06	2012
12th	U.P BOARD PRAYAGRAJ U.P	U.P BOARD,UTTAR PRADESH	70.02	2006
10th	U.P BOARD PRAYAGRAJ U.P	U.P BOARD,UTTAR PRADESH	66.83	2004

Operating Systems : Windows and Linux

Internship-6 month training with Sandeepani school of Embeded system design Bangalore cours E on FPGA DESIGN-VLSI112 & DIGITAL SYSTEM DESIGN USING VERILOG IEEE Certification VLSI INDUSTRY.(2022-23)

Additional Information

Title of paper: Published Research Papers:

"An Efficient approach of True Random Number Generation using Serial Input Parallel Output Register & Tree based Interleaver"

International Journal of Engineering Research & Technology (IJERT)

ISSN: 2278 – 0181, **DOI : 10.17577/IJERTV11IS070007**, VOLUME 11, ISSUE 07 (JULY 2022)

"A Novel Approach of True Random Number Generation Using a Vedic Multiplier "

Conference Publications/JOURNAL PUBLICATION:

[1]:A Novel Approach of true random number generation using a vedic multiplier.journal of pharmaceutical negative results|VOLUME 13|SPECIAL ISSUSE7|2022

Certification of Presentation of 41st World Conference On Applied science , Engineering & Technology(WCASET 2022) on 24 & 25 August virtual conference "

ISSN:41-WCASET-2022_ETH_206, PUBLISHED IN SCOPUS INDEX JOURNAL ,

VOL.ISSUSE 07(AUG 2022).

[2]: An Efficient Analysis for True Random Number Generators using A Postprocessor, Jitter Metastability, Tree-Based Interleaver & CMOS inverter prototype technology .(ICAEEET2022)

[3]:AN Efficient approach of True random Number Generation using A DATA Encryption Methoed in security & cryptography algorthim [ICICAT 2023] PUBLISHED IN IEEE EXPLORE 2023 PROCEEDING.

AREAS OF INTEREST: Embeded system & Computer Architecture Organisation , Basic Electronics Theory ,Signal and Systems | Communication Systems | ElectronDevices and Circuit Theory | Control System | Integrated MicroElectronics CircuitAnalog Design|Microwave & sattalite Communication|Optical Fiber communication| Electronics & Digital Logic Electronics | VLSI Design |Electromagnets &Electronics Instrumentation control,Optical & Sattalite Communication system, Artificial Intelligence for Internet of Things,Data science & Machine Learning using Numpy, R-systemStudio,Pandas,Matplotlib, Data Lab view simulation using Multisim 14.02

MANAGEMENT SUBJECTS: Finance management,Finance Accounting for managers, Cyber security, HR& Marketing Business Strategic,Economics,Micro & Macro Economics Management ,Indian Economics manament.Management Function Behavioural information system , Basics of Economics Fundaments ,Research Methodology,DATA SCIENCE Analysis.

Enterprise Managementn,information security, Risk & Production Total quality Management ,Humam & professional Ethics ,Buisness organization behavior,sales & Digital Marketing. Quantitative Research Operation Management.Management FunctionsSystem Analyst. Information.Quantitative & Legal Aspect Techniques, Project computer Application mana-Gement.Corporate Tax ,consumer & supply chain Management,management Functions & Behaviour ,

List of International/National Conferences/Symposia Attended:

1. **Workshops:**

Organised & attended a workshop titled Workshop On Particals On Matlab organised by Department Of Electronics Engineering, IETE Lucknow, April 08- 04, 2008

2. Organised & attended a workshop titled Practicle Workshop On Embeded System Embeded System Design Based On Microelectrocontrollers (Phoenix) organised by Department Of Electronics Engineering,V.I.E.T Gearter Noida, Oct 28, 2011 .

2.Seminars & Conferences Attended: 1.National Conference On Advancement in Satellite Communication (14-15); conducted at Ajay Kumar Garg Engineering College Ghaziabad (Akgec College Code-027) Uttar Pradesh

2. One Day National Seminar On September 7, 2013 titled Advances In Power System (Nsaps-2013) organised By Department Of Electrical Engineering in Kamla Nehru Institute Of Engineering & Technology, Sultanpur (KNIT), Uttar Pradesh

3.Three Day 3rd International Conference On Microelectronics Devices Circuits and system 11th to13 august 2022 Organised By Department Of

Micro & Nano Electronics School of Electronics Engineering Springer & Global Foundaries VIT VELLORE ICMDCS2022 Tamilnadu

4. Certificate of Participation Two Day International Conference On Trends in Electronics and Health Informatics,TEHI-21 Organised By Pranveer Singh Institute Of Technology , KANPUR Held online During Dec 16-17,2021
5. Certification Of Participation Five Days National Workshop on Data Science and Machine Learning (NWDSML-22) Organised by Department of Mathmatics school of Basic Sciences, C.S.J.M University KANPUR U.P
6. Second International Conference(virtual) on Artificial Intelligence for Internet of Things AIIOT-2022 , BY Institutions Innovation council , SENSE ,UTeM(University Teknikal Malaysia) Department of Embeded Technology & School Of Electronics Eng. ,MHRD held at VIT VELLORE TAMILNADU(07-08 April 2022.
7. Certificate of Participation a Three Day CSIR sponsored Workshop on " Recents Trends in Green Energy Drives" With Hands on Training by School of Electrical Eng. VIT VELLORE from 07/04/2022 to 09.04.2022
8. .Certificate of Participation & coordinator counselor Of Five Day FDP-cum-STTP on Simulation , Modelling and Application of Advanced Semiconductor Devices from 05-09 July ,2021 Organised by Department of Electronics Eng. Harcourt Butler Technical University Kanpur-208002
9. Certificate of Participation online E& ICT FDP On " Data science & Machine Learning from Aug 26 , 2022 To sep 04-2022, Organised by Department of CSE , NIT Warangal 506004, Telangana and School of Information Tech and Eng (SITE), Vit Vellore , Tamilnadu
10. Certificate of Intended Participants a Three –Day Workshop on ElectromagneticInterference and Compatibility For Industrial and Medical Aplication on dated21st-23rd July 2022 Dr. Jameel Akhtar, Professor Deparment of Electrical Engineering IIT KANPUR.

Awards & Achievements

Received **Certificate of Merit** in Robotics competition held in Cognizence Festival in IIT Roorkee.

Actively participated in various talent search examinations at school and college level

Participated in a Blood Donation held at V.I.E.T Greater Noida

Participated as a Team member in Techkriti Festival in INDIAN INSTITUTE TECHNOLOGY Kanpur

- **Strengths** :1. Achieved a Qualifying Gate Score in Graduate Aptitude Test in Engineering in 2012-13
- **Achieved a opportunity Giving a UGC National Eligibility Test in Electronics science Subject for Lectureship UGC NTA Score In academic year 2017,2018,2019,2020,2021**

,

Declaration

I do hereby confirm that the information given above is true and to the best of my knowledge.

Prateek Agnihotri Kanpur Uttar Pradesh

Your Name

CURRICULUM VITAE

ER.PRATEEK AGNIHOTRI

Senior FPGA /Verification/Electronics Eng./Senior FPGA VLSI SYSTEM EMBEDED CLOUD IOT, Hardware-software Eng.

DOMAIN-DIGITAL RTL DESIGN,DESIGN VERIFICATION,UVM ,SYSTEM VERILOG,AXI,UART .LOW POWER CMOS VERIFICATION SPI,MICROPROCESSOR,PIC MICROCONTROLLER VERIFICATION ,Debug,PCB , Software Tools, Simulation based verification Icauras verilog

Designation: Digital RTL Eng.Trainee, Design Verification , VLSI PHYSICAL DESIGN ,DFT DESIGN,RTL DESIGN,DIGITAL DESIGN ENG, ANALOG DESIGN USING CMOS ,EMBEDED SYSTEM,ARM PROCESSOR,MICROCONTROLLER,PIC ,8086 MICROPROCEESOR CIRCUIT SEMICONDUCTOR DESIGN TRAINEE ENGINEER

Departmen Senior FPGA /Verification/Electronics Eng./Senior FPGA VLSI SYSTEM EMBEDED CLOUD IOT, Hardware-software Eng

1. **DOMAIN-DIGITAL RTL DESIGN,DESIGN VERIFICATION,UVM ,SYSTEM VERILOG,AXI,UART .LOW POWER CMOS VERIFICATION SPI,MICROPROCESSOR,PIC MICROCONTROLLER VERIFICATION ,Debug,PCB , Software Tools, Knowledge of EDA tools, Chip design, LINUX etc.,PHYSICAL DESIGN WITH STA ,LRC,TAPOUT CHIP**
2. **LAYOUT CHIP DESIGN FABRICATION,VERIFICATION,PHYSICAL SYNTHESIS**

Building IoT applications using Arduino and Raspberry pi,

Internship on Embedded System Design and IoT (30 Day's)

Internship on FPGA DIGITAL RTL TO GDS FLOW DESIGN (UNDER MINISTRY OF EDUCATION pantechsolutions.net/

Internship on Embedded System Design and IOT

Internship on Embedded C Programming

Internship on Internship on ESD & IoT using ARM Cortex M4 FPGA

:FOR Electronics& Communication Eng. ,Electrical & Electronics Eng,M.B.A,B.B.A

Qualification :B.TECH[Electronics & Telecommunication Eng.],M.TECH[VLSI &

EMBEDDED SYSTEM DESIGN],
M B A – information Technology FROM AMITY UNIVERSITY NOIDA U.P GLOBAL U.S CERTIFIED ,Data Science Advanced certification under iit madaras
With Intellipatt sol.pvt ltd Banhalore –IT,AI,ML ,Data science Advanced Technology.

Phone: 7991613936,9451218659 Email: agnihotriprateek.88@gmail.com,

Skill Set

OBJECTIVE: To associate with multi-dynamic institute, which may provide a platform to update my knowledge & skills in accordance with latest trends in teaching along with research follows a tradition of anticipating & leading changes

Proficiency: (Change or Add more as necessary)

HDL	Hardware	Programming languages	Tools
Verilog	ZedBoard	C (Basic),VHDL,PYTHON,VERILOG,SYSTEM VERILOG	QuestaSim, Vivado,MATLAB,,SPICE,MySQL,Xilinx ISE9.2
Technical Skills and Tools	Programming Language : C, Verilog and System Verilog.	Scripting Language : TCL.,Perl script	Tools : Synopsys Sentaurus TCAD, Xilinx Vivado ,Questa Sim Xilinx Ise 9.2 design suit,Mentor graphics, Aimspice ,pspice ,silvaco Microwind, Synopsys Prime time STA Analysis, DFT Altera ,synopsys DC Compiler,,Modelsim,Questa sim For Digital design Verification ,Ncsim ,VHDL,Cadence Design Virtosoo,IR ,RTL LINT Jasper,

Operating Systems : Windows and Linux
Internship-6 month training with Sandeepani school of Embeded system designBangaluru cours E on FPGA DESIGN-VLSI112 & DIGITAL SYSTEM DESIGN USING VERILOG IEEE Certification VLSI INDUSTRY.(2022-23) , I

hv also work upon system verilog Fundamental &Verified Risc , Arm processor so attached o

- Certified UNDER " MAVEN SILICON Bangaluru" & “NIELIET NPTEL CERIFICATION **& CHIP EDGE TECHNOLOGIES PVT.LTD, BTM LAYOUT,BANGALURU ,KARNATAKA 560076.**

Reference /Reference-By Manikantha vaka and all team work organisation

Experience

- Worked with **Bharat Electronics Limited** Ghaziabad from June, 2012till Dec, 2012 as Trainee Engineer [contract position]incuding internet of things,arudino, PIC ,with & RF Microwave Digital Propagation .(Devices wave propagation).
- Worked With BHABHA College of Science & Tech . Kanpur Assistant Prof. /Research Assistant and with Digital & Analog communication & Microwave ,Anteena RF Design subject with industry 4.0 B.tech Course structure .
- **Worked with Bhabha Institute of Science and Technology[BIST] ,MANAGEMENT ,Technical VLSI EMBEDED & MICROPROCESSOR FPGA MODEL SIM, VIVADO HLS,QUESTA SIM FPGA SYNTHESIS DESIGN MANUAL (VLSI ARM ,DIGITAL VLSI EMBEDED SYSTEM DESIGN-112 LAB ASSISTANT in ECE DEPARTMENT TECHNICAL,TRAINER IN Kanpur Dehat from July 2012to june 2018.**
- **Demonstrated experience working on FPGA design projects, including work with SoC (ARM/RISC-V CPU), 10/ 40/ 100G Ethernet PHY, DDR#, PCI-E, SDIO interface, & Sandeepani school of Embeded system designBangalore,as Physical design digital FPGA Module design ENGINEER AS Level-I UPTO Senior VLSI FPGA DESIGN Eng.& verification including SPI.UART,FIFO,ARM ,SOC AMBA BUS Ethernet, Bus architecture incuding 8086 microcontroller & Pheripheral interface devices domain ,10 hb receiving Ethernet module design (Tx & Rx both designed).**

.

Major Projects

- **Project 1**: **Project** – Functional Specification for SPI IP PROTOCOL Design
  March 2022to Aug 2022

Master mode and slave mode

• Bi-directional mode • Slave select output • Serial clock with programmable polarity and phase. Clock Polarity and Clock Phase:Data Transmission .

- **MOSI:** MOSI stands for Master Output Slave Input. It is used to send data from the master to the slave.
- **MISO:** MISO stands for Master Input Slave Output. It is used to send data from the slave to the master.
- **SCK or SCLK (Serial Clock):** It is used to the clock signal.
- **SS/CS (Slave Select / Chip Select):** It is used by the master to send data by selecting a slave.

Note: If the single slave is present in the communication, that required only three wires. The SS (slave select) is not required in in it.

a. **Project 2 – Designing of To**

Generate a True Rndom Number Generator using vedic multiplier, ring oscillator circuit XOR Gate , Tetra Hedral Ring Oscillation LFSR IDMA AND COMPARE CALIBRATION GENERATION METHOED USING VHDL SIMULATION

SEP,2021 to SEP, 2022-Final Thesis Submiteed under guidance of Prof. Dr.Ashok kumar shankwar & co guidance joint by dr, preeti agarwal mittal & Rajni bisht(Retired HBTU KANPUR)
TITLE:An Efficient Approach of True Random number generation using A seed Ring oscillation & Tree-based Interleaver.in Xilinx Verilog,VIVADO 2020.20,Aim spice,LT spice , MATLAB , MODEL SIM -Cryptography technoiques encryption,decryption.

we propose a new method for generation of TRNG using tree based interleaving technique. The output of one oscillator is send to SIPO register using a XOR gate. The other input of XOR gate is kept at '1'. The property of XOR gate is that the output bit will be independent if its input bits are independent. After the SIPO register the output is connected to the XOR based postprocessor. Among Von Neuman, BCH code and XOR based preprocessor we have used XOR based preprocessor due to ease of implementation. Finally, for implementing tree based interleaving multiplexer is used with its select lines connected to binary up down counter.

Minor Projects

- **Mini project 1: Desighning a 10 gb Ethernet Card june 2011to aug 2011[Thesis submiteed Bachlor of Tech. VLSI DESIGN Under Guidence of Prof. M.C Bansal (Retired IIT DELHI ,IIT BHU under guidance)Thesis:Design of Reciver module of 10 GB Ethernet card ,TX-RX Ethernet design system Application.**

- Client: Ducat Noida
- Location: Noida
- Project: Design a 10 gb Ethernet card
- Tools & Technology: VHDL, VERILOG SYSTEM
-
- **Project Description:**
- The Ethernet specification served as the basis for the IEEE 802.3 standard, which specifies the physical and lower software layer. -Worked on Networking and Telecommnication
-Ethernet and SDH technologies
-Ethernet MAC(AXI data stream interface) and PHY interfaces(RGMII/GMII)
-10Mbps to 100Gbps Ethernet
-Ethernet Protocols(IP/UDP, ARP/RARP, PING)
-MDIO, AXI4-lite, I2C, SPI interfaces
-SDH-POS/ATM
-DDR2/DDR4
-Good knowledge on CDC
-Vivado
-Simulation using Vivado
-timing analysis
-On-board testing and debug

Mini project 2: Project – Designing of Microcontroller 8085,8086
Feb,2010 to Mar, 2011

Board Level Testing and Validation
Architecture & RTL Design
FPGA Design and Implementation
Board Design and Layout
Board Assembly and Manufacturing
Embedded and Software Engineering
Board Bring Up, Testing and Validation

SoC Design and Verification with key focus on Processor Subsystem design (UVM)
IP Integration and Verification
RTL Design Simulation
Gate Level Sign Off,
Place and Route
Physical Verification
Package Design and and Verification,
Architecture
Simulation and ATPG

SOC Design & Signoff
Low Power Design

- Client: DUCAT Noida
- Location: Noida
- Project: Designing of Microcontroller 8085,8086
- Tools & Technology: Embedded System, VHDL, VERILOG
- **Project Description:**
- The 8085 incorporates the functions of the 8224 (clock generator) and the 8228 (system controller), increasing the level of integration. A downside compared to similar contemporary designs (such as the Z80) was the fact that the buses required de-multiplexing

Educational Qualifications

Course	Institution	University / Board	Aggregate Percentage	Batch
M.Tech. (VLSI Design & Embedded System)	HARCOURT BUTLER TECHNICAL UNIVERSITY KANPUR	H.B.T.U KANPUR UTTAR PRADESH	70.02	2023
M.BA (IINFORMATION TECHNOLOGY)	AMITY UNIVERSITY SECTOR-125 Noida UTTAR PRADESH-201303	AMITY UNIVERSITY ODL ,Noida Uttar Pradesh	65.06	2021-23
B.Tech (Electronics & Communication)	VISHVESHWARYA INSTITUTEOF ENG. & Tech. G. B NAGAR GR. NOIDAU.P	Dr.APJ ABDUL KALAM TECHNICAL UNIVERSITY LUCKNOW U.P	65.06	2012
12th	U.P BOARD PRAYAGRAJ U.P	U.P BOARD,UTTAR PRADESH	70.02	2006
10th	U.P BOARD PRAYAGRAJ U.P	U.P BOARD,UTTAR PRADESH	66.83	2004

Additional Information

Title of paper: Published Research Papers:

"An Efficient approach of True Random Number Generation using Serial Input Parallel Output Register & Tree based Interleaver"

International Journal of Engineering Research & Technology (IJERT)

ISSN: 2278 – 0181, **DOI : 10.17577/IJERTV11IS070007**, VOLUME 11, ISSUE 07 (JULY 2022)

"A Novel Approach of True Random Number Generation Using a Vedic Multiplier "

Conference Publications/JOURNAL PUBLICATION:

[1]:A Novel Approach of true random number generation using a vedic multiplier.journal of pharmaceutical negative results|VOLUME 13|SPECIAL ISSUSE7|2022

Certification of Presentation of 41st World Conference On Applied science , Engineering & Technology(WCASET 2022) on 24 & 25 August virtual conference "

ISSN:41-WCASET-2022_ETH_206, PUBLISHED IN SCOPUS INDEX JOURNAL ,

VOL.ISSUSE 07(AUG 2022).

[2]: An Efficient Analysis for True Random Number Generators using A Postprocessor, Jitter Metastability, Tree-Based Interleaver & CMOS inverter prototype technology .(ICAEEET2022)

[3]:AN Efficient approach of True random Number Generation using A DATA Encryption Methoed in security & cryptography algorthim [ICICAT 2023] PUBLISHED IN IEEE EXPLORE 2023 PROCEEDING.

AREAS OF INTEREST: Signal and Systems | Communication Systems | Electron Devices and Circuit Theory | Control System | Integrated Analog Electronics & Digital Logic Electronics | VLSI Design |Electromagnets & Electronics Instrumentation control,Optical & Sattalite Communication system, Artificial Intelligence for Internet of Things,Data science & Machine Learning using Numpy, R-systemStudio,Pandas,Matplotlib, Data Lab view simulation using Multisim 14.02,IT MANAGEMENT, HR MANAGEMENT.FINANCE MANAGEMENT, CRM , BUISNESS MANAGEMENT,MANAGEMENT FUNCTION, BEHAVIOURL MANAGEMENT,CRM RELATIONSHIP MANAGEMENT, IT DATABASE ADMINISTRATION (along with Intellipatt sol.Ltd IIT MADRAS CERTIFIED Training certificate)

Certification Courses

• ARM Based SOC Design By Nieliet Calicut

• VLSI FUNDAMENTALS BY Nieliet Calicut,IOT FUNDAMENTAL BY NIELIET CALICUT

SYSTEM VERILOG VERIFICATION BY MAVEN SILICON BANGLORE KARNATAKA.

Area of Interest

Semiconductor Physics

Digital System Design,FPGA, verilog verification,VHDL verification

VLSI physical Design

Digital IC Design.

List of International/National Conferences/Symposia Attended:

3. **Workshops:**

Organised & attended a workshop titled Workshop On Particals On Matlab organised by Department Of Electronics Engineering, IETE Lucknow, April 08- 04, 2008

4. Organised & attended a workshop titled Practicle Workshop On Embeded System Embeded System Design Based On Microelectrocontrollers (Phoenix) organised by Department Of Electronics Engineering,V.I.E.T Gearter Noida, Oct 28, 2011 .

2.Seminars & Conferences Attended: 1.National Conference On Advancement in Satellite Communication (14-15); conducted at Ajay Kumar Garg Engineering College Ghaziabad (Akgec College Code-027) Uttar Pradesh

8. One Day National Seminar On September 7, 2013 titled Advances In Power System (Nsaps-2013) organised By Department Of Electrical Engineering in Kamla Nehru Institute Of Engineering & Technology, Sultanpur (KNIT), Uttar Pradesh

9.Three Day 3rd International Conference On Microelectronics Devices Circuits and system 11th to13 august 2022 Organised By Department of Micro & Nano Electronics School of Electronics Engineering Springer & Global Foundaries VIT VELLORE ICMDCS2022 Tamilnadu

10. Certificate of Participation Two Day International Conference On Trends in Electronics and Health Informatics,TEHI-21 Organised By Pranveer Singh Institute Of Technology , KANPUR Held online During Dec 16-17,2021

11. Certification Of Participation Five Days National Workshop on Data Science and Machine Learning (NWDSML-22) Organised by Department of Mathmatics school of Basic Sciences, C.S.J.M University KANPUR U.P

12. Second International Conference(virtual) on Artificial Intelligence for Internet of Things AIIOT-2022 , BY Institutions Innovation council , SENSE ,UTeM(University Teknikal Malaysia) Department of Embeded Technology & School Of Electronics Eng. ,MHRD held at VIT VELLORE TAMILNADU(07-08 April 2022.

13. Certificate of Participation a Three Day CSIR sponsored Workshop on " Recents Trends in Green Energy Drives" With Hands on Training by School of Electrical Eng. VIT VELLORE from 07/04/2022 to 09.04.2022

8..Certificate of Participation & coordinator counselor Of Five Day FDP-cum-STTP on Simulation , Modelling and Application of Advanced Semiconductor Devices from 05-09 July ,2021 Organised by Department of Electronics Eng. Harcourt Butler Technical University Kanpur-208002

11. Certificate of Participation online E& ICT FDP On " Data science

& Machine Learning from Aug 26 , 2022 To sep 04-2022, Organised by Department of CSE , NIT Warangal 506004, Telangana and School of Information Tech and Eng (SITE), Vit Vellore , Tamilnadu

12. Certificate of Intended Participants a Three –Day Workshop on ElectromagneticInterference and Compatibility For Industrial and Medical Aplication on dated21st-23rd July 2022 Dr. Jameel Akhtar, Professor Deparment of Electrical Engineering IIT KANPUR.

Awards & Achievements

Received **Certificate of Merit** in Robotics competition held in Cognizence Festival in IIT Roorkee.

Actively participated in various talent search examinations at school and college level

Participated in a Blood Donation held at V.I.E.T Greater Noida

Participated as a Team member in Techkriti Festival in INDIAN INSTITUTE TECHNOLOGY Kanpur

- **Strengths** :1. Achieved a Qualifying Gate Score in Graduate Aptitude Test in Engineering in 2012-13

- . Physical design engineer supporting the full chip creation, layout and final delivery of GDS-2.FPGA DEBUG,VIVADO SYNTHESIS,OPTIMIZATION,SIMULATION ,STA ,CTS,ROUTING,PIN PLACEMENT , FLOORPLANING,IP DEBUG , LUT SYNTHESIS, using LINUX,TCL compiler commands Lab Manuals, PHYSICAL VERIFICATION (SYNOPSYS DESIGN COMPILER,LVS,DRC, TIMING ANALYSIS, SIGNOFF Analysis IN FPGA DIGITAL DESIGN VERIFICATION ,IR DROP ,LEC,CLOCK TREE SYNTHESIS,LVS,SIGN-OFF,Design Rule Violation check , Physical Design , DFT Mannual, Physical design verification ,DFT , Macros, STA AT Different Phases ,
-
- Work with ASIC vendor to drive them for timing closure/STA, layout, RAS features and on chip repair methodology for the best solution in the chip industry.
-
- Develop and own physical design implementation of multi-hierarchy low-power designs including physical-aware logic synthesis, design for testability, floorplan, place and route, static timing analysis, IR Drop, EM, and physical verification in advanced technology nodes.
-
- Resolve design and flow issues related to physical design, identify potential solutions, and drive execution.
-
- Deliver multiple ASICs with physical design of an end-to-end integration of ASIC/SoC design.
- Varrious Linux operating system environment software courses, Project Worked upon based on that & multiple IT related capston projects, including , PHYTHON, ML ,R-SYSTEM, SQL,AZURE,POWER BI,JUPYTER NOTE BOOK, Microsoft EXEL Core Data set,Time series ,statistical Analysis using SQL ,Artificial Intelligence.

  Kaggle data set experiment capstone projects .

  Experience in scan insertion, coverage analysis and debugging skills on faults coverage enhancement is required.
  · Programming in Perl/Tcl/Python or other scripting languages is a plus.
  · Experience in post silicon validation, ATE debug and support is desired.
  · Exposure to RTL2GDS flow and tasks such as synthesis and scan insertion, STA and IR drop.
  · Good understanding of Logic design, RTL implementation & verification, logic synthesis, Logic Equivalent Checking & Static Timing Analysis are plus.

· Participate in driving new DFT methodology and solutions to improve quality, reliability and in-system test and debug capability, hands-on experience using industry standard EDA Tools.
· Experience with logic simulators from one or more EDA vendors.
· Experience on industry standard ATPG tools like Cadence Modus.
· Experience with RTL lint tools like Jasper

Involve in design and system architecture for complex FPGA based Test solutions Develop Verilog code for various system blocksParticipate in size/ timing optimization of complete FPGA design
Place & route flows, pinning, and timing optimization for Xilinx and/ or Altera Continuously improving test and simulation environment to assure FPGAdesign work

programming of 8051 series base nuvoton mcu.

Design and implement the software for embedded devices and systems, from creation to production and deployment.

STM 32cube MCU PACKAGES,STM32C0,STM32F0,STM32F1

https://www.linkedin.com/in/prateek-agnihotri-20136341/

	CADENCE	SYNOPSYS	MENTOR GRAPHICS (SIEMENS)
Synthesis	GENUS	DESIGN COMPILER	PRECISION SYNTHESIS
Placement and Route	INNOVUS	SYNOPSYS ICC ICC2 SHELL	OLYMPUS-SoCINNOVUS
Physical Design Verification (DRC, LVS)	CADENCE PVS	IC VALIDATOR	CALIBRE
RC Extraction	QUANTUS	STAR-RCXT	CALIBRE RC
Static Time Analysis (STA)	TEMPUS	PRIMETIME-PX	CALIBRE SST-Velocity
Power Analysis	CADENCE VOLTUS	PRIME POWER	
Simulation	CADENCE XSIM	SYNOPSYS VCS	ModelSim

Declaration

I do hereby confirm that the information given above is true and to the best of my knowledge.

Prateek Agnihotri
25/10/2023
Your Name